
Image Filtering and Histogram

Do the following

1. Write a code to apply the following kernels.

(a)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \frac{1}{9}$$

(b)

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

(c)

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

2. Define a function that takes an image and kernel as an argument and return the resultant image.
3. Do the following.

- (a) Derivate in x and y .
- (b) Find the Magnitude M.
- (c) Threshold the image.

Note: The submitted assignment should consist of a screen shot on a page showing the code and output result.